🍔 SWIGGY SALES ANALYSIS - BUSINESS PROBLEM STATEMENT

🎯 EXECUTIVE SUMMARY

Swiggy, one of India's leading food delivery platforms, operates in a highly competitive market with over 1,000 restaurant partners across 10 major cities. Despite achieving ₹39.61 Crores in revenue over the past 4 years (2022-2025), the company faces critical challenges in optimizing operations, improving customer retention, and maximizing profitability.

This comprehensive data analysis project aims to uncover actionable insights from 600,000+ orders to drive strategic decision-making across multiple business functions.

📋 BACKGROUND

Company Overview

Swiggy is a food ordering and delivery platform connecting customers with restaurants across India. The platform operates on a commission-based model, earning revenue from:

- Restaurant commissions (15-25% per order)
- Delivery fees charged to customers
- Premium subscription services
- Advertising and promotional placements

Current Situation

Over the past 4 years, Swiggy has:

- Served **10,000+ active customers**
- Partnered with **1,000 restaurants** across 10 cities
- Processed **600,000 orders**
- Generated **₹39.61 Crores** in gross order value
- Maintained a **95% order fulfillment rate**

Despite this growth, several challenges have emerged that require data-driven solutions.

🔴 KEY BUSINESS CHALLENGES

1. CUSTOMER RETENTION & ENGAGEMENT

Problem:

- High customer acquisition costs (CAC) are eating into profit margins
- Customer churn rate is unknown and potentially high

- Lack of understanding about customer lifetime value (CLV)
- No clear segmentation strategy for targeted marketing

Questions to Answer:

- What is our customer retention rate?
- Who are our most valuable customers?
- How can we identify customers at risk of churning?
- What drives customer loyalty and repeat purchases?
- Which customer segments should we prioritize?

2. REVENUE OPTIMIZATION

Problem:

- Revenue growth has plateaued in certain months/cities
- Unclear understanding of revenue drivers
- Seasonal patterns not being leveraged effectively
- Pricing strategy lacks data backing

Questions to Answer:

- What are the key revenue drivers (city, cuisine, time, customer segment)?
- Which months/quarters show the highest revenue? Why?
- What is the year-over-year growth rate?
- How does average order value (AOV) vary across segments?
- What pricing strategies can maximize revenue without losing customers?

3. RESTAURANT PARTNER PERFORMANCE

Problem:

- Significant variance in restaurant performance
- Underperforming restaurants affecting platform reputation
- Unclear criteria for onboarding new restaurant partners
- No systematic approach to restaurant optimization

Questions to Answer:

- Which restaurants generate the most revenue?
- How does restaurant rating impact order volume?
- What is the performance difference between cloud kitchens and traditional restaurants?
- Which cuisines are most popular in which cities?
- Which restaurants should we prioritize for marketing investments?
- Which underperforming restaurants need support or should be removed?

4. MENU & PRODUCT OPTIMIZATION

Problem:

- No clear understanding of menu item performance
- Vegetarian vs non-vegetarian preference patterns unknown
- Pricing strategy for different categories not optimized
- Cross-selling and upselling opportunities not identified

Questions to Answer:

- What are the best-selling menu items overall and by category?
- How does vegetarian vs non-vegetarian preference vary by city/demographic?
- What is the optimal price range for different menu categories?
- Which items are frequently ordered together (basket analysis)?
- What menu items drive the highest margins?

5. OPERATIONAL EFFICIENCY

Problem:

- 5% order cancellation rate causing revenue loss and customer dissatisfaction
- Suboptimal delivery capacity during peak hours
- Resource allocation not aligned with demand patterns
- Unclear understanding of operational bottlenecks

Questions to Answer:

- What is the hourly/daily order distribution pattern?
- When are the peak ordering times? Are we adequately staffed?
- What is the order cancellation rate by city/time/restaurant?
- Why are orders being cancelled?
- How can we optimize delivery operations?
- What is the weekend vs weekday demand difference?

6. MARKET EXPANSION & GROWTH

Problem:

- Unclear which cities/areas to prioritize for expansion
- Market saturation in some cities vs untapped potential in others
- No data-driven approach to restaurant onboarding
- Competitive landscape not well understood

Questions to Answer:

- Which cities show the highest revenue per restaurant?
- Which cities have room for more restaurant partners?

- What is the optimal restaurant density per city?
- Which cuisines are underrepresented in which cities?
- Where should we expand next?

7. CUSTOMER EXPERIENCE & SATISFACTION

Problem:

- No systematic analysis of customer behavior patterns
- Payment method preferences not being leveraged
- Demographic preferences not understood
- Personalization opportunities not identified

Questions to Answer:

- What are the preferred payment methods by demographic?
- How do different age groups/occupations behave differently?
- What factors influence order value?
- How can we personalize the customer experience?
- What drives customer satisfaction vs dissatisfaction?

🎯 PROJECT OBJECTIVES

Primary Objectives

1. **Develop Customer Segmentation Strategy**
   - Identify high-value customer segments
   - Create RFM (Recency, Frequency, Monetary) segmentation
   - Calculate customer lifetime value
   - Develop targeted retention strategies
2. **Optimize Revenue Generation**
   - Identify top revenue drivers
   - Develop pricing optimization recommendations
   - Create seasonal marketing strategies
   - Maximize average order value
3. **Improve Operational Efficiency**
   - Reduce order cancellation rate
   - Optimize delivery capacity allocation
   - Improve demand forecasting
   - Streamline restaurant operations
4. **Enhance Restaurant Partner Ecosystem**
   - Identify high-performing restaurant characteristics
   - Develop restaurant performance scoring
   - Create restaurant onboarding criteria
   - Optimize restaurant mix by city
5. **Drive Strategic Growth**
   - Identify expansion opportunities
   - Develop market penetration strategies

- Create competitive positioning
- Enable data-driven decision making

📊 EXPECTED DELIVERABLES

1. Analytical Deliverables

- **Customer Analysis Report**
  - Customer segmentation (Champions, Loyal, At-Risk, Lost)
  - Lifetime value analysis
  - Churn prediction and prevention strategies
  - Demographic behavior patterns
- **Revenue Analysis Report**
  - Revenue trends and forecasts
  - Key revenue drivers identification
  - Seasonal pattern analysis
  - Pricing optimization recommendations
- **Restaurant Performance Report**
  - Top and bottom performing restaurants
  - Rating impact analysis
  - Cuisine performance by city
  - Cloud kitchen vs traditional comparison
- **Menu & Product Analysis Report**
  - Best-selling items analysis
  - Vegetarian vs non-vegetarian insights
  - Cross-selling opportunities
  - Price elasticity analysis
- **Operational Analysis Report**
  - Peak hours and demand patterns
  - Cancellation root cause analysis
  - Capacity optimization recommendations
  - Delivery efficiency metrics

2. Visual Deliverables

- **Interactive Power BI Dashboard** :
  - Executive summary page
  - Customer analytics page
  - Restaurant performance page
  - Product insights page
  - Operational metrics page
  - City-wise performance page

3. Strategic Recommendations

- **Customer Retention Strategy**
  - Loyalty program design

    - Re-engagement campaigns
    - Personalization roadmap
- **Revenue Growth Strategy**
    - Pricing optimization
    - Cross-selling initiatives
    - Seasonal campaigns
- **Operational Excellence Roadmap**
    - Capacity planning model
    - Cancellation reduction plan
    - Process improvements
- **Expansion Strategy**
    - City prioritization
    - Restaurant onboarding criteria
    - Market entry recommendations

🔍 ANALYTICAL APPROACH

Data Analysis Methodology

1. **Descriptive Analytics** (What happened?)
    - Historical trend analysis
    - Performance metrics calculation
    - Distribution analysis
2. **Diagnostic Analytics** (Why did it happen?)
    - Root cause analysis
    - Correlation analysis
    - Comparative analysis
3. **Predictive Analytics** (What will happen?)
    - Customer churn prediction
    - Revenue forecasting
    - Demand prediction
4. **Prescriptive Analytics** (What should we do?)
    - Optimization recommendations
    - Strategic recommendations
    - Action plans

Tools & Technologies

- **Database**: PostgreSQL
- **Analysis**: Python (Pandas, NumPy)
- **Visualization**: Power BI, Matplotlib, Seaborn
- **Documentation**: Jupyter Notebooks, Markdown